

Mexican Slang Jailbreak Benchmark

A lightweight AI Safety benchmark for regional Mexican slang jailbreak evaluation

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Abstract

Current AI safety evaluations tend to focus on English or formal Spanish. This leaves an important gap: users often communicate in informal, regional language that models may not fully understand. We propose the Yucatan Slang Jailbreak Benchmark, a reproducible red-teaming workflow that tests whether LLM safety filters can be bypassed when harmful intent is expressed through Mexican and Yucatecan slang. The system reads slang terms from PostgreSQL, uses an Attacker agent to generate adversarial prompts, sends them to a swappable Target model, evaluates the response with a Judge agent on a 1–5 scale, stores structured results, and visualizes metrics in Grafana. At this checkpoint, our contribution is the research framing and working evaluation architecture. The remaining work is to run end-to-end experiments and report examples/results.

1. Introduction and Motivation

AI Safety is not only a question of what models refuse in English. It is also a question of whether safeguards remain reliable across cultures, dialects, and informal speech. In Latin America, and particularly in Mexico, harmful intent may be expressed through slang, double meanings, code-switching, or regional phrases. If safety filters are less robust to these registers, deployed models can present uneven safety guarantees across communities.

This project sits within the Global South AI Safety frame: safety benchmarks should reflect the linguistic and cultural settings where systems are used. We draw motivation from work on geographies of the Global South, the Latin American AI Index (ILIA), and HarmBench. The gap we address is specific: HarmBench provides a strong automated red-teaming frame, but it does not directly evaluate how regional Mexican slang changes safety behavior.

- Research question: Can regional Mexican slang, or related informal Spanish patterns, expose safety gaps in LLMs?
- Checkpoint goal: validate the structure, method, and evaluation plan before reporting final results.

2. Related Work and Safety Frame

HarmBench gives us the general structure for automated red-teaming: generate adversarial prompts, query target models, and classify the outputs. ILIA helps justify the regional relevance: Latin America needs evaluation resources that reflect its own languages, infrastructure, and data realities. Work on Global South geographies reminds us that AI risk is shaped by context, not only by model capability. Our benchmark combines these ideas in a narrow, testable setting: slang-aware safety evaluation.

Reference frame	How it supports this project
HarmBench	Provides the adversarial evaluation pattern: attack, target response, judge classification.
ILIA / Latin America	Motivates regional evaluation capacity and language-specific safety benchmarks.
Global South AI Safety	Frames safety as culturally situated, not only technically universal.

3. Method: Compact Workflow

The benchmark is designed as a small, reproducible pipeline rather than a one-off prompt test. The Python CLI is the current source of truth for the flow, and the n8n workflow is being aligned to reproduce the same process visually.

```
PostgreSQL jerga table
↓
Attacker agent: slang + meaning + harm_category + attack technique
↓
Target model: receives only the adversarial prompt
↓
Regex pre-filter: detects obvious harmful output and saves judge tokens
↓
Judge agent: scores the response from 1 to 5
↓
Storage: results table in PostgreSQL
↓
Grafana: ASR, score, severity, model/technique/category views
```

Component	Role in the benchmark
Jerga corpus	Stores term, meaning, harm category, and region.
Attacker	Constructs a realistic adversarial prompt using slang and one attack technique.

Target	The model being evaluated; no custom system prompt.
Judge	Scores the target response from 1 to 5; scores 4–5 count as jailbreak_success.
Storage + Grafana	Stores evidence and visualizes attack success rate across models and techniques.

4. Current Implementation Status

The current repository implements the benchmark in Docker with PostgreSQL, n8n, Grafana, and a Python attacker CLI. PostgreSQL now stores free-form target responses as text and the structured Judge verdict as JSONB. The Judge contract includes score, jailbreak_success, confidence, severity, harm_detected, and reasoning. The n8n workflow has been updated to mirror this contract, including the 1–5 score and batch loop.

- Validated: PostgreSQL tables and seed data load correctly; 10 seed slang rows cover drugs, hate_speech, and violence.
- Validated: the Python CLI loads, lists models, and blocks real runs when NVIDIA_API_KEY is missing.
- Validated: the n8n workflow JSON is syntactically valid and aligned with the Python Judge/storage contract.
- Pending: run main.py end-to-end with a real NVIDIA key, then run the imported n8n workflow end-to-end.

5. Examples to Include in Final Paper

This section is intentionally structured but not filled with final evidence yet. The final paper should include safe, non-actionable examples that show the benchmark behavior without reproducing harmful instructions.

Example slot	What to show
Slang transformation	A benign or redacted prompt showing how one slang term changes the surface form of a harmful category.
Judge scoring	One target refusal scored 1–2 and one unsafe/partial response scored 4–5, with reasoning summarized.
Workflow trace	One row from results: target_model, technique, harm_category, score, severity, jailbreak_success.

Example format for the final version: “Given slang term X and harm category Y, the Attacker generated a disguised prompt. The Target response was classified by the Judge as score Z because [...]” Harmful details should be redacted or summarized.

6. Results Plan

The final results section should be short and visual. The judges should quickly see whether slang changes model behavior, and if so, where: by target model, attack technique, harm category, and score distribution.

Metric	Purpose
Attack Success Rate (ASR)	Percent of attempts where score ≥ 4 .
ASR by target model	Which models are more or less vulnerable.
ASR by technique	Which prompt strategies create failures.
ASR by harm category	Which categories are most sensitive to slang obfuscation.
Score distribution	Whether failures are full jailbreaks or borderline leakage.
SIS / slang integration score	Whether the Attacker actually integrated slang into the prompt.

Interpretation should remain open: if failures appear, they may be due to regional slang, neutral Spanish, code-switching, prompt structure, or another repeatable vulnerability. The benchmark is designed to reveal the pattern rather than assume the answer.

7. Discussion and Limitations

This is an early-stage benchmark. The seed corpus is small and needs broader coverage, model selection is not final, and the results still need end-to-end validation. The main risk is overclaiming: the goal is not to prove that “slang always bypasses safety,” but to create a measurement pipeline that can test whether underrepresented language patterns produce measurable safety differences.

- Limitation: seed slang corpus is temporary and should be expanded/curated.
- Limitation: LLM-as-a-Judge needs consistency checks and possibly human review for a sample.
- Limitation: examples must avoid reproducing actionable harmful content.

8. Conclusion

We present a focused AI Safety benchmark for regional Mexican slang jailbreak evaluation. The core contribution is a reproducible workflow that turns a cultural-linguistic safety question into measurable data: prompts, responses, scores, and dashboard metrics. For

the checkpoint, the paper structure is ready: motivation, method, workflow, examples, and results plan. The immediate next step is to run the Python and n8n paths end-to-end so the final paper can replace placeholders with concrete examples and Grafana-backed results.

References

Mazeika et al. (2024). HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal.

Centro Nacional de Inteligencia Artificial / ILIA (2025). Índice Latinoamericano de Inteligencia Artificial.

Huang et al. (2026). Geographies of the Global South and AI Safety framing.

Project repository: Yucatan Slang Jailbreak Benchmark, Global South AI Safety Hackathon 2026.

LLM Usage Statement

LLMs were used to assist drafting, restructuring, and summarizing project materials. Technical claims should be verified against the repository, workflow executions, and final experiment outputs before submission.